

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5113

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Annexure A

1. Event Communication Security (Caesar Cipher, Shift = 4)

Original message:

MEET AT NOON

A Caesar cipher with a shift of 4 moves each letter four positions forward in the alphabet.

Plaintext M E E T A T N O O N

Ciphertext Q I I X E X R S S R

Encrypted message:

QIIX EX RSSR

How an authorized member decrypts it

Since the member knows the shift is 4, they move each ciphertext letter 4 positions backward:

- $Q \rightarrow M$
- $I \rightarrow E$
- $I \rightarrow E$
- $X \rightarrow T$

Repeating this for all letters reconstructs:

MEET AT NOON

2. Intercepted Suspicious Message

Ciphertext:

GSRH RH Z HVXIVG

This strongly resembles an **Atbash cipher**, a substitution cipher where:

- $A \leftrightarrow Z$
- $B \leftrightarrow Y$
- $C \leftrightarrow X$
- ...
- $M \leftrightarrow N$

Decoding

Cipher G S R H

Plain T H I S

So:

- $GSRH \rightarrow THIS$
- $RH \rightarrow IS$
- $Z \rightarrow A$
- $HVXIVG \rightarrow SECRET$

Original message:

THIS IS A SECRET

Justification

1. The word pattern GSRH frequently appears as THIS in Atbash examples.
2. The one-letter word Z maps naturally to A.
3. The final word length and letter structure decode perfectly to SECRET.
4. Atbash is a simple monoalphabetic substitution cipher, matching the scenario description.

3. Secure Messaging System (Vigenère Cipher)

Plaintext: HELLO

Keyword: KEY

Step 1: Repeat the keyword

Plaintext: H E L L O

Keyword: K E Y K E

Step 2: Convert letters to positions

Letter Value

A	0
B	1
...	...
H	7
E	4
L	11
O	14
K	10
Y	24

Step 3: Add plaintext and keyword values (mod 26)

Plain	H E L L O
Value	7 4 11 11 14
Key	K E Y K E
Key Value	10 4 24 10 4
Sum mod 26	17 8 9 21 18
Cipher	R I J V S

Ciphertext

RIJVS

Influence of the repeating keyword

The keyword determines the shift for each character:

- K causes a shift of 10
- E causes a shift of 4
- Y causes a shift of 24

Because the keyword repeats (KEYKE...), different letters in the message are shifted by different amounts, making frequency analysis more difficult than with a simple Caesar cipher.

4. Suspicious Digital Asset Investigation (Steganography)

How investigators can detect hidden data

Technique 1: Statistical Analysis

Investigators examine pixel values and image characteristics for anomalies.

Examples:

- Unusual patterns in least significant bits (LSBs)
- Unexpected color distributions
- Abnormal image entropy

Such irregularities may indicate embedded information.

Technique 2: Metadata and File Analysis

Investigators inspect:

- File size inconsistencies
- Hidden metadata
- Extra data appended after image end markers
- Differences between original and suspect versions

Specialized steganalysis tools can reveal concealed payloads.

Two techniques to strengthen hidden communications

1. Encrypt the data before embedding

Even if hidden data is discovered, strong encryption (e.g., AES) prevents an attacker from reading the content.

2. Use adaptive or randomized embedding

Instead of inserting data in predictable pixel locations:

- Randomize embedding positions using a secret key.
- Embed only in noisy or textured regions.

This makes statistical detection and extraction much harder.

5. Legacy Encryption in Corporate Communication

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

This is a classic **Rail Fence Cipher (3 rails)** example.

(a) Recovering the original message

Arrange the ciphertext into three rails and reconstruct the zigzag pattern.

The decrypted message is:

WE ARE DISCOVERED FLEE AT ONCE

(Without spaces: WEAREDISCOVEREDFLEEATONCE)

(b) Why this method is vulnerable today

1. **No secret key complexity**
 - Only the number of rails needs to be guessed.
2. **Preserves letter frequencies**
 - The letters themselves are unchanged.
 - Frequency analysis remains effective.
3. **Small key space**
 - Attackers can quickly test all plausible rail counts.
4. **Easily broken by computers**
 - Modern tools can brute-force transposition ciphers almost instantly.

Therefore, Rail Fence encryption provides very weak protection against modern attacks.

(c) Practical enhancement

A practical improvement is to **encrypt the message with a modern symmetric algorithm such as AES before transmission.**

Benefits:

- Strong resistance to brute-force attacks.
- Secure key-based encryption.
- Protects message content even if intercepted.

For organizations requiring confidentiality, modern cryptographic standards (AES with proper key management) are vastly more secure than classical transposition methods.

Summary

Scenario	Method	Result
Event communication	Caesar Cipher (+4)	MEET AT NOON → QIIX EX RSSR
Intercepted message	Atbash Cipher	GSRH RH Z HVXIVG → THIS IS A SECRET
Keyword encryption	Vigenère Cipher	HELLO + KEY → RIJVS
Hidden image data	Steganography	Detect via statistical and metadata analysis
Legacy communication	Rail Fence (3 rails)	WECRLTEERDSOEFEAOCAIVDEN → WE ARE DISCOVERED FLEE AT ONCE